

Pankaj Thakur

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PROFESSIONAL SUMMARY

Full-Stack & AI Engineer (B.Tech AI & DS, MITS Gwalior, 2027) · GATE CSE 2026 **Top 7.3% nationally** (AIR 14,658 / 200K+ candidates). Ships production RAG, multi-agent AI, and SaaS systems using FastAPI, LangGraph, React, Node.js, MongoDB, and Groq.

EDUCATION

Madhav Institute of Technology & Science (MITS), Gwalior

Sep 2023 – Aug 2027

B.Tech in Artificial Intelligence & Data Science

CGPA: 8.3 / 10

Relevant Coursework: DSA, DBMS, OOP, Operating Systems, Machine Learning, NLP, Computer Networks

EXPERIENCE

Full Stack Developer Intern

May 2026 – Jun 2026

CodeAlpha (Remote)

- Architected and shipped a full-stack Content Management Platform with JWT authentication, RBAC, 8 RESTful API endpoints, and Zod middleware validation; delivered zero runtime errors to production within a 6-week sprint

TECHNICAL SKILLS

- Languages:** Python, C++, JavaScript (ES6+), SQL, HTML5, CSS3
- Frontend:** React.js, Vite, Tailwind CSS, React Router DOM, Streamlit, Leaflet
- Backend:** Node.js, Express.js, FastAPI, REST API Design, JWT Authentication, RBAC, Zod
- Databases:** MongoDB Atlas, SQLite, MySQL, ChromaDB (Vector Store)
- AI / GenAI:** LangChain, LangGraph, RAG, Groq, OpenAI Whisper, Prompt Engineering, Agentic AI, Multi-Agent Systems, NLP
- Tools:** Git, GitHub, Docker, Postman, Razorpay, Google OAuth, Socket.io, SSE, Tavily API, FFmpeg, pytest, CI/CD
- Deployment:** Vercel, Render, MongoDB Atlas, Streamlit Cloud

PROJECTS

Summly — AI Meeting Intelligence Platform [\[GitHub\]](#)

May 2026

- Designed 5-stage ML audio pipeline (FFmpeg → spectral denoising → Whisper int8 → LangGraph → Groq) auto-extracting meeting action items, decisions, and follow-ups; surpassing Whisper baseline transcription accuracy by 8% on a 30-recording evaluation set.
- Implemented production RAG pipeline over meeting transcripts using sentence-transformer embeddings + ChromaDB (top-3 retrieval); achieved sub-3s query latency and 95%+ answer relevance (RAGAS-evaluated) across a 50-session test corpus.
- Shipped 20+ JWT-secured REST APIs (bcrypt, YouTube ingestion, AI-generated email follow-ups) with 80% pytest coverage, enabling CI/CD-ready zero-downtime deployment on Render.

SmartPark — Full-Stack Parking SaaS [\[GitHub\]](#) [\[Live Demo\]](#)

Apr 2026

- Architected multi-tenant SaaS with 3-tier RBAC (Driver/Owner/Admin) using MongoDB ACID transactions and pessimistic locking; validated zero data conflicts across 100+ simulated concurrent booking requests.
- Engineered 23 production-hardened REST APIs with JWT auth, express-rate-limit throttling, and XSS/NoSQL sanitization; integrated Socket.io delivering real-time slot updates with sub-100ms event propagation.
- Developed multi-factor geo-ranking algorithm via MongoDB \$geoNear aggregation; secured payments with Razorpay HMAC-SHA256 webhook verification and Google OAuth 2.0, achieving near-zero payment failure rate in production.

ResearchOS — Multi-Modal AI Research Platform [\[GitHub\]](#) | [\[Live Demo\]](#)

Mar 2026

- Architected multi-modal AI research platform with 21 JWT-secured REST APIs, reducing end-to-end research synthesis from 4+ hours to under 5 minutes — a 12× speed improvement validated across 25+ test queries.
- Designed 4-stage autonomous LLM agent pipeline (Search → Read → Write → Review) using LangGraph + Groq with async SSE streaming at ~500 tok/s; auto-generates fully cited reports with quality scoring in under 5 minutes.
- Implemented session-isolated RAG pipeline (HuggingFace embeddings + ChromaDB) with LangChain multi-tool agent integrating Tavily API and weather data for context-augmented research generation.

ACHIEVEMENTS & CERTIFICATIONS

- GATE CSE 2026** — **Top 7.3% nationally** (AIR 14,658 among 200,000+ candidates), Score: 426.
- LeetCode [\[Profile\]](#)** — 120+ DSA problems solved across arrays, trees, graphs, and DP.
- Certifications:** [PW Skills DSA](#) | [Udemy Backend Development](#)